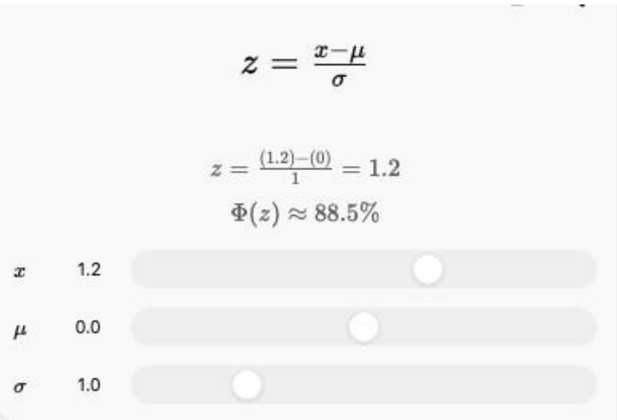
C0-2 AT-3
Case study question

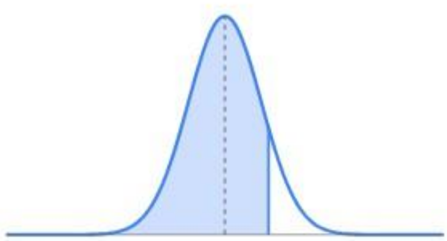
Case Study 6 — Detecting an Unusual Customer

A shopping website finds that the average amount spent by customers is **₹2,000**, with a standard deviation of **₹400**.

A particular customer spends **₹3,200**.

1. Calculate the customer's **z-score**.
2. Is the customer's spending unusually high compared with the average?
3. What would a negative z-score mean in this context?
4. If another customer has a z-score of −1.5, what does it tell you?





Solution:

Given Data

- Mean (μ) = ₹2,000
- Standard Deviation (σ) = ₹400
- Customer Spending (x) = ₹3,200

Question 1: Calculate the Customer's Z-Score
Formula

$$z = \frac{x - \mu}{\sigma}$$

where

- x= Customer's spending
- μ= Mean spending
- σ= Standard deviation

Step 1: Write the given values

$$x = 3200,\ \mu = 2000,\ \sigma = 40$$

Step 2: Substitute the values

$$z = \frac{3200 - 2000}{400}$$

Step 3: Subtract the numerator

$$3200 - 2000 = 1200$$
$$z = \frac{1200}{400}$$

Step 4: Divide

$$1200 \div 400 = 3$$

Answer

$$\boxed{z = 3}$$

The customer's spending is 3 standard deviations above the average.

Question 2: Is the Customer's Spending Unusually High Compared with the Average?
A z-score helps us understand how unusual a value is.

- If z = 0, the value is exactly equal to the average.
- If z = 1, it is one standard deviation above the average.
- If z = 2 or more, it is considered unusually high.
- Here, z = 3.

Answer
Yes. The customer's spending is unusually high because the z-score is 3, which means the customer spends 3 standard deviations above the average.

Question 3: What Would a Negative Z-Score Mean in This Context?
A negative z-score means the customer's spending is less than the average spending.
For example:

- z = -1 means the customer spends 1 standard deviation below the average.
- z = -2 means the customer spends much less than the average.

Answer
A negative z-score indicates that the customer spends below the average amount.

Question 4: If Another Customer Has a Z-Score of –1.5, What Does It Tell You?
A z-score of –1.5 means the customer spends 1.5 standard deviations below the average.
This customer spends less than the average customer, but it does not necessarily mean the spending is abnormal.
Answer
A customer with a z-score of –1.5 spends 1.5 standard deviations below the average, indicating lower-than-average spending.

Case Study 7 — Delivery Time Analysis

A delivery company records delivery times (in minutes):

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

1. Calculate the mean and median.
2. Find Q1 and Q3.
3. Calculate the IQR.
4. Check for outliers using the 1.5 × IQR rule.
5. Explain whether the mean or median gives a better description of the typical delivery time.

Solution:

Question 1: Calculate the Mean and Median

Step 1: Count the observations

There are **10** delivery times.

$$n = 10$$

Step 2: Find the sum

$$25 + 27 + 28 + 29 + 30 + 30 + 31 + 32 + 34 + 60 = 326$$

Step 3: Apply the formula

$$\text{Mean} = \frac{326}{10} = 32.6$$

Answer

Mean = 32.6 minutes

Median

Step 1

The data is already arranged in ascending order.

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

Step 2

There are 10 values (even number).

So the median is the average of the 5th and 6th values.

5th value = 30

6th value = 30

Step 3

$$\text{Median} = \frac{30 + 30}{2} = 30$$

Answer

Median = 30 minutes

Question 2: Find Q1 and Q3

Step 1

Divide the data into two equal halves.

Lower half:

25, 27, 28, 29, 30

Upper half:

30, 31, 32, 34, 60

Step 2: Find Q1

The middle value of the lower half is

25, 27, **28**, 29, 30

$$Q_1 = 28$$

Step 3: Find Q3

The middle value of the upper half is

30, 31, **32**, 34, 60

$$Q_3 = 32$$

Answer

$$Q_1 = 28,\ Q_3 = 32$$

Question 3: Calculate the IQR

Formula

$$IQR = Q_3 - Q_1$$

Step 1

Substitute the values.

$$IQR = 32 - 28$$

Step 2

Subtract.

$$IQR = 4$$

Answer

IQR = 4

Question 4: Check for Outliers Using the 1.5 × IQR Rule

Step 1: Find 1.5 × IQR

$$1.5 \times 4 = 6$$

Step 2: Lower Limit

$$Q_1 - 1.5(IQR)$$
$$28 - 6 = 22$$

Lower Limit = **22**

Step 3: Upper Limit

$$Q_3 + 1.5(IQR)$$
$$32 + 6 = 38$$

Upper Limit = **38**

Step 4: Compare the Data

Value	**Result**
25	Within limit
27	Within limit
28	Within limit
29	Within limit
30	Within limit
30	Within limit
31	Within limit
32	Within limit
34	Within limit
60	Above 38 → Outlier

Answer

60 is an outlier because it is greater than the upper limit (38).

Question 5: Explain whether the mean or median gives a better description of the typical delivery time. (Short Answer)

The median gives a better description of the typical delivery time because the value 60 minutes is an outlier. The outlier increases the mean (32.6 minutes), while the median (30 minutes) remains unaffected and better represents the typical delivery time.

Case Study 8 — Two Cities' Temperatures

The average daily temperatures (°C) for 7 days are:

City A: 28, 29, 30, 30, 31, 30, 32
City B: 20, 25, 30, 35, 30, 25, 45

1. Calculate the mean temperature for each city.
2. Calculate the variance for each city.
3. Which city has greater temperature variability?
4. Why can two datasets have the same mean but different standard deviations?
5. Which city has more predictable temperatures?

Solution:

Given Data

The average daily temperatures (°C) for 7 days are:

City A: 28, 29, 30, 30, 31, 30, 32

City B: 20, 25, 30, 35, 30, 25, 45

Question 1: Calculate the Mean Temperature for Each City

Formula

$$\text{Mean} = \frac{\text{Sum of all observations}}{\text{Number of observations}}$$

City A

Step 1: Count the observations

The temperatures are:

28, 29, 30, 30, 31, 30, 32

There are **7 observations**.

$$n = 7$$

Step 2: Find the sum

Add all the temperatures one by one.

$$
\begin{gathered}
28 + 29 = 57 \\
57 + 30 = 87 \\
87 + 30 = 117 \\
117 + 31 = 148 \\
148 + 30 = 178 \\
178 + 32 = 210
\end{gathered}
$$

Therefore,

$$\text{Sum} = 210$$

Step 3: Apply the formula

$$\text{Mean} = \frac{210}{7}$$

Step 4: Divide

$$210 \div 7 = 30$$

Answer

Mean Temperature of City A = 30°C

City B

Step 1: Count the observations

20, 25, 30, 35, 30, 25, 45

There are **7 observations**.

Step 2: Find the sum

$$
\begin{aligned}
&20 + 25 = 45 \\
&45 + 30 = 75 \\
&75 + 35 = 110 \\
&110 + 30 = 140 \\
&140 + 25 = 165 \\
&165 + 45 = 210
\end{aligned}
$$

Therefore,

$$\text{Sum} = 210$$

Step 3: Apply the formula

$$\text{Mean} = \frac{210}{7}$$

Step 4: Divide

$$210 \div 7 = 30$$

Answer

Mean Temperature of City B = 30°C

Question 2: Calculate the Variance for Each City

Formula

$$\text{Variance} = \frac{\sum(x - \mu)^2}{n}$$

where:

- x= each temperature
- μ= mean
- n= number of observations

City A

Mean = **30°C**

Temperature (x)	$x - 30$	$(x - 30)^2$
28	-2	4
29	-1	1
30	0	0
30	0	0
31	1	1
30	0	0
32	2	4

Sum of squared deviations

$$4 + 1 + 0 + 0 + 1 + 0 + 4 = 10$$

Variance

$$\frac{10}{7} = 1.43$$

Answer

Variance of City A = 1.43

City B

Mean = **30°C**

Temperature (x)	$x - 30$	$(x - 30)^2$
20	-10	100
25	-5	25
30	0	0
35	5	25
30	0	0
25	-5	25
45	15	225

Sum of squared deviations

$$100 + 25 + 0 + 25 + 0 + 25 + 225 = 400$$

Variance

$$\frac{400}{7} = 57.14$$

Answer

Variance of City B = 57.14

Question 3: Which City Has Greater Temperature Variability?

Temperature variability means how much the temperatures change from the average.

- City A Variance = 1.43
- City B Variance = 57.14

Since 57.14 > 1.43, City B has greater temperature variability.

Answer

City B has greater temperature variability because its variance is much higher than City A's.

Question 4: Why Can Two Datasets Have the Same Mean but Different Standard Deviations?

The mean tells us the average value of a dataset, while the standard deviation measures how spread out the values are around the mean.

In this case:

- Both City A and City B have the same mean (30°C).
- However, City A's temperatures are close to 30°C every day.
- City B's temperatures range from 20°C to 45°C, so they are much more spread out.

Answer (Short)

Both cities have the same average temperature (30°C), but City B's temperatures vary much more from the average. Therefore, City B has a larger standard deviation.

Question 5: Which City Has More Predictable Temperatures?

A city with less variation has more predictable temperatures.

- City A has a very small variance (1.43).
- City B has a much larger variance (57.14).

Therefore, City A's temperatures are more consistent from day to day.

Answer (Short)

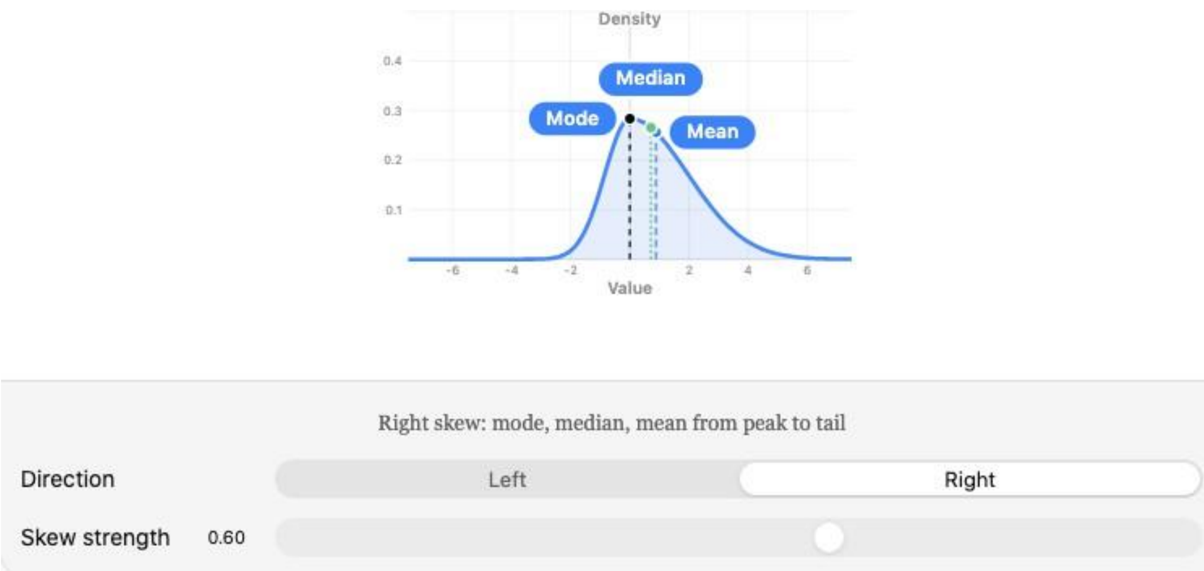
City A has more predictable temperatures because its temperatures stay close to the average, resulting in much lower variability.

Case Study 9 — Data Center CPU Usage

A data scientist records CPU usage percentages:

40, 42, 45, 45, 46, 48, 50, 52, 55, 90

1. Calculate mean, median, and mode.
2. Calculate the range.
3. Find Q1, Q3 and IQR.
4. Identify possible outliers.
5. Determine the direction of skewness.



Solution:

Given Data
CPU usage percentages:
40, 42, 45, 45, 46, 48, 50, 52, 55, 90
The data is already arranged in ascending order.

Question 1: Calculate Mean, Median, and Mode
(A) Mean
Formula

$$\text{Mean} = \frac{\text{Sum of all observations}}{\text{Number of observations}}$$

Step 1: Count the observations
There are 10 CPU usage values.

$$n = 10$$

Step 2: Find the sum

$$40 + 42 + 45 + 45 + 46 + 48 + 50 + 52 + 55 + 90$$

Adding step by step,

$$40 + 42 = 82$$
$$82 + 45 = 127$$
$$127 + 45 = 172$$
$$172 + 46 = 218$$
$$218 + 48 = 266$$
$$266 + 50 = 316$$
$$316 + 52 = 368$$
$$368 + 55 = 423$$
$$423 + 90 = 513$$

Therefore,

$$\text{Sum} = 513$$

Step 3: Divide

$$\text{Mean} = \frac{513}{10} = 51.3$$

Answer
Mean = 51.3%

(B) Median
Since there are 10 values (even number),
Median = Average of the 5th and 6th values.
5th value = 46
6th value = 48

$$\text{Median} = \frac{46 + 48}{2}$$
$$= \frac{94}{2} = 47$$

Answer
Median = 47%

(C) Mode
The mode is the value that occurs most frequently.
40, 42, 45, 45, 46, 48, 50, 52, 55, 90
Only 45 appears twice.
Answer
Mode = 45%

Question 2: Calculate the Range
Formula

$$\text{Range} = \text{Maximum Value} - \text{Minimum Value}$$

Step 1: Find the maximum value
Largest value = 90
Step 2: Find the minimum value
Smallest value = 40
Step 3: Subtract

$$90 - 40 = 50$$

Answer
Range = 50

Question 3: Find Q1, Q3 and IQR
Step 1: Divide the data into two halves
Lower Half
40, 42, 45, 45, 46
Upper Half
48, 50, 52, 55, 90

Step 2: Find Q1
Middle value of lower half
40, 42, **45**, 45, 46

$$Q_1 = 45$$

Step 3: Find Q3
Middle value of upper half
48, 50, **52**, 55, 90

$$Q_3 = 52$$

Step 4: Calculate IQR
Formula

$$IQR = Q_3 - Q_1$$

Substitute the values

$$IQR = 52 - 45$$
$$IQR = 7$$

Answer
- Q1 = 45
- Q3 = 52
- IQR = 7

Question 4: Identify Possible Outliers
Use the 1.5 × IQR Rule
Step 1: Calculate 1.5 × IQR

$$1.5 \times 7 = 10.5$$

Step 2: Lower Limit

$$Q_1 - 1.5(IQR)$$
$$45 - 10.5 = 34.5$$

Step 3: Upper Limit

$$Q_3 + 1.5(IQR)$$
$$52 + 10.5 = 62.5$$

Step 4: Compare the values

Value	Result
40	Within limit
42	Within limit
45	Within limit
45	Within limit
46	Within limit
48	Within limit
50	Within limit
52	Within limit
55	Within limit
90	Above 62.5 → Outlier

Answer

90 is the possible outlier because it is greater than the upper limit (62.5).

Question 5: Determine the Direction of Skewness

To determine skewness, compare the mean and median.

- Mean = 51.3
- Median = 47

Since

$$\text{Mean} > \text{Median}$$

the distribution is right-skewed (positively skewed).

This happens because the high value 90 pulls the mean towards the right.

Answer (Short)

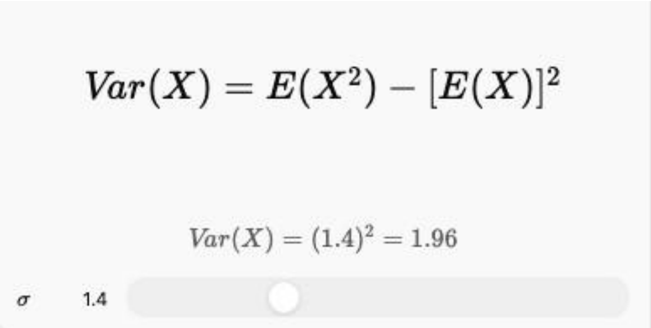
The data is right-skewed (positively skewed) because the outlier 90 increases the mean above the median.

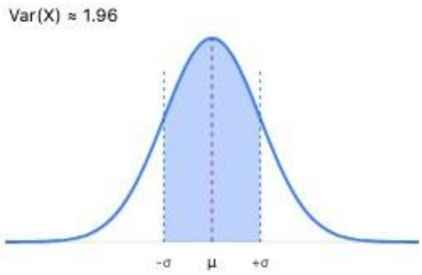
Case Study 10 — Comparing Two Datasets

Two data analysts receive the following datasets:

Dataset A: 10, 20, 30, 40, 50
Dataset B: 28, 29, 30, 31, 32

1. Calculate the mean of both datasets.
2. Calculate the variance of both.
3. Calculate the standard deviation of both.
4. Which dataset has greater spread?
5. If both datasets represent model errors, which model appears more consistent?
6. Explain why **mean alone is not enough** to compare datasets.





Solution:

Given Data
Dataset A
10, 20, 30, 40, 50
Dataset B
28, 29, 30, 31, 32

Question 1: Calculate the Mean of Both Datasets
Formula

$$\text{Mean} = \frac{\text{Sum of observations}}{\text{Number of observations}}$$

Dataset A
Step 1: Count the observations
There are **5 values**.

$$n = 5$$

Step 2: Find the sum

$$10 + 20 + 30 + 40 + 50 = 150$$

Step 3: Divide

$$\text{Mean} = \frac{150}{5} = 30$$

Answer
Mean of Dataset A = 30

Dataset B
Step 1: Count the observations
There are 5 values.
Step 2: Find the sum

$$28 + 29 + 30 + 31 + 32 = 150$$

Step 3: Divide

$$\text{Mean} = \frac{150}{5} = 30$$

Answer
Mean of Dataset B = 30

Question 2: Calculate the Variance of Both Datasets
Formula

$$\text{Variance} = \frac{\sum(x - \mu)^2}{n}$$

where:

- x= each value
- μ= mean
- n= number of observations

Dataset A
Mean = **30**

x	**x−30**	**(x−30)²**
10	-20	400
20	-10	100
30	0	0
40	10	100
50	20	400

Sum of squared deviations

$$400 + 100 + 0 + 100 + 400 = 1000$$

Variance

$$\frac{1000}{5} = 200$$

Answer
Variance of Dataset A = 200

Dataset B
Mean = **30**

x	x−30	(x−30)²
28	-2	4
29	-1	1
30	0	0
31	1	1
32	2	4

Sum

$$4 + 1 + 0 + 1 + 4 = 10$$

Variance

$$\frac{10}{5} = 2$$

Answer
Variance of Dataset B = 2

Question 3: Calculate the Standard Deviation of Both
Formula

$$\text{Standard Deviation} = \sqrt{\text{Variance}}$$

Dataset A
Variance = **200**

$$\sqrt{200} = 14.14$$

Answer
Standard Deviation = 14.14

Dataset B
Variance = 2

$$\sqrt{2} = 1.41$$

Answer
Standard Deviation = 1.41

Question 4: Which Dataset Has Greater Spread?
The spread of a dataset tells us how far the values are from the mean.

- Dataset A has Variance = 200 and Standard Deviation = 14.14.
- Dataset B has Variance = 2 and Standard Deviation = 1.41.

Since Dataset A has much larger variance and standard deviation, its values are more spread out.
Answer
Dataset A has the greater spread because its values are much farther from the mean.

Question 5: If Both Datasets Represent Model Errors, Which Model Appears More Consistent?
A model is more consistent when its errors are close to each other, meaning it has a smaller standard deviation.

- Dataset A: SD = 14.14
- Dataset B: SD = 1.41

Dataset B has the smaller standard deviation.
Answer
Dataset B appears more consistent because its prediction errors are closer to the mean and show less variation.

Question 6: Explain Why Mean Alone Is Not Enough to Compare Datasets
Both datasets have the same mean (30).
However:

- Dataset A values are spread widely (10 to 50).
- Dataset B values are close together (28 to 32).

This shows that the mean only tells the average. It does not show how the values are distributed. Measures like variance and standard deviation are needed to understand the spread of the data.
Answer (Short)
Mean alone is not enough because two datasets can have the same average but very different spreads. Variance and standard deviation help measure the variability of the data.